

Single-Channel Speech Separation

The influence of additive noise on blind source separation

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- ① Task Description
- ② Choice of noise values
- ③ Training Strategy
- ④ Results
- ⑤ section5

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Task Description

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- Analyse the performance of a separation model as the signal-to-noise ratio (SNR) decreases.
- Input: Mixture signal with noise of varying levels
- Measure the discrepancy between the estimated sources and the true sources

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- Additive White Gaussian Noise (AWGN), $\epsilon \sim \mathcal{N}(0, \sigma^2)$.
- To simulate varied acoustic interference levels

Chosen sigma values:

- $\sigma = 0.005 \implies \text{SNR} \approx 17.4 \text{ dB}$
- $\sigma = 0.01 \implies \text{SNR} \approx 11.3 \text{ dB}$
- $\sigma = 0.02 \implies \text{SNR} \approx 5.3 \text{ dB}$

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2 Choice of noise values

3 Training Strategy

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4 Results

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Table of Results

Model	$\sigma = 0.005 (SNR = 17.4dB)$	$\sigma = 0.01 (SNR = 11.3dB)$	$\sigma = 0.02 (SNR = 5.3dB)$
Original model	-4.22	-3.62	-1.86
Trained on σ_2	-4.50	-4.21	-3.13

表 1: Results - SI-SNR-PIT on test set

Mixed Signal

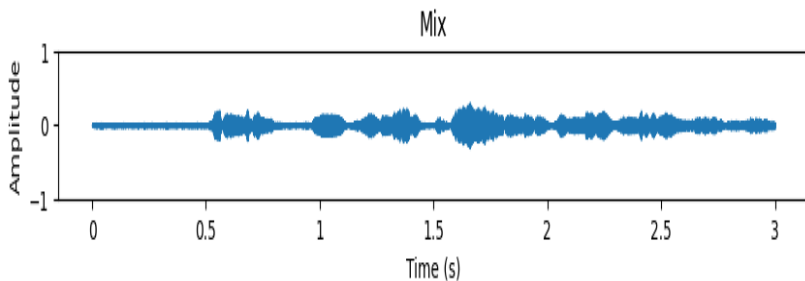
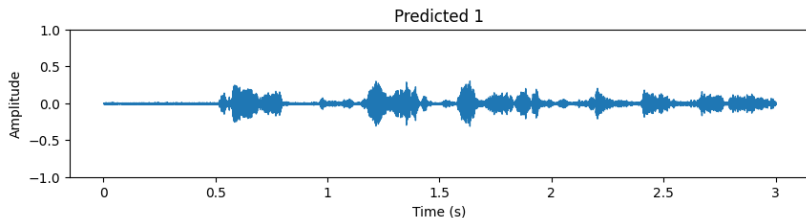
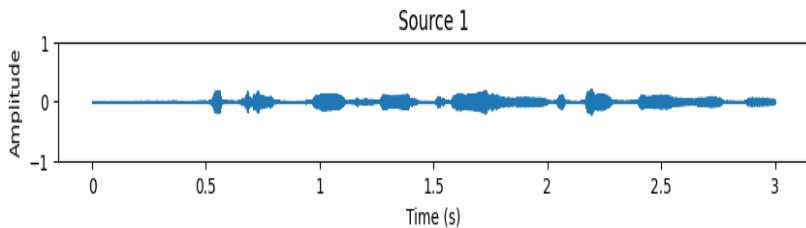
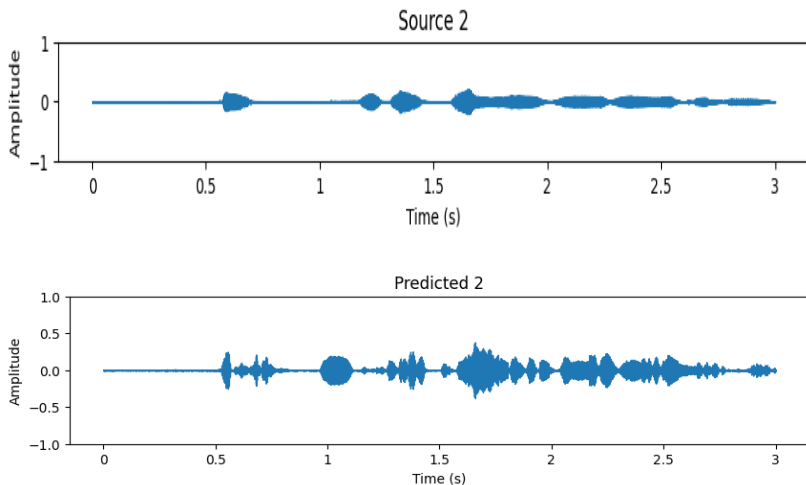


图 1: Plot of Mixed Signals

First Source



Second Source



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Thanks!